

leading to a total privacy budget of $(16, 2e^{-5})$ due to the sequential composition law of the DP mechanism (Abadi et al., 2016). These settings align with established DP deployments such as Apple’s Quick-Type and Google’s models, as noted by Lukas et al. (2023). More comparisons between our method and baselines under different privacy budgets are presented in Section 6.6.

During synthetic data sampling, we use the vLLM framework (Kwon et al., 2023) for fast inference, setting the batch size to 32 and sampling 6 candidate responses for each synthetic query. The sampling templates are detailed in Appendix J. For Reward Guided Filtering, we sort the dataset by reward score, split it into k folds, and select the fold with the highest score, setting k to 6 for medical QA, 5 for financial QA, and 8 for code generation. For Self-Optimizing Refinement, we set the number of candidate responses N as 3 for medical QA and code generation, 2 for financial QA task. The hyperparameter analysis is provided in Section 6.6 and Appendix I. The generation temperature is 1.0 and top-p is 0.7 to enhance diversity. The templates used for generating feedback are provided in Appendix J.

For LLM fine-tuning on the server, we use the standard SGD algorithm and train the model for 3 epochs with a learning rate of $4e-5$ and a batch size of 64. The maximum sequence length for all fine-tuning processes is set to 768. All training and generation processes are conducted on an A800 80G.

E Details of LLM-Judge Evaluation

Considering ROUGE-L/ROUGE-1 metrics only measure lexical similarity to references and PPL only captures fluency, they often fail to assess deeper aspects of response quality. To ensure more reliable evaluation on the generated outputs for the medical QA and financial QA tasks, we adopt the LLM-Judge approach (Zheng et al., 2023) for assessment.

First, we fine-tune the LLM-Judger for these domain-specific tasks (medical QA and financial QA). The fine-tuning process is similar to that of our reward proxy model, where we construct preference pair data as training data and use Bradley-Terry loss (Liu et al., 2024) for training. The key difference is that we use the more powerful Qwen2.5-13B-Instruct backbone and fine-tune it with the AdamW optimizer, without adding DP

noise. We fine-tune the LLM-Judger for 3 epochs with a learning rate of $4e-5$.

During evaluation, we provide the LLM-Judger with both the user query and the generated output, allowing the judger to score the outputs. The judge template is provided in Appendix J. We then compare the scores of outputs from our method and other baselines. If the score difference is less than 1, it is considered a tie. Otherwise, the output with the higher score is viewed as the winner.

As shown in Figure 3, our method outperforms other baselines in both the medical QA and financial QA tasks. DP-Generation and KnowledgeSG struggle with noisy samples from synthetic data, leading to poor performance. Although DP-Instruct filters synthetic data by comparing with private data and removing low-similarity samples, it achieves only limited performance gains compared to DP-Generation. This shows that simple similarity measures do not fully capture the quality of synthetic data. Locally Fine-tuning avoids noise from synthetic data by fine-tuning a lightweight proxy model on private data locally, but it still underperforms our method due to the limited learning capacity of the lightweight model for domain-specific knowledge.

F Case studies

Here, we present a representative example to demonstrate the effectiveness of our method by comparing its generated response with those from baseline methods, including DP-Generation, DP-Instruct, and KnowledgeSG.

As shown in Figure 7, **DP-Generation** includes repetitive and irrelevant symptoms, such as no facial weakness and no difficulty swallowing, which are not directly related to the user’s query. **DP-Instruct** avoids repeating unrelated symptoms but still offers unhelpful advice, only suggesting that the user see a doctor without providing any meaningful medical analysis. Similarly, **KnowledgeSG** offers some advice, like conducting a physical examination, but also fails to provide any professional analysis of the user’s symptoms or potential underlying causes.

In contrast, **RewardDS** provides a more detailed analysis of the user’s symptoms, offers some possible causes, and suggests feasible advice, such as scheduling a cardiologist appointment and undergoing an ECG test, strongly demonstrating its effectiveness.



User Query: I was sitting in class when my pulse suddenly started to accelerate for no apparent reason, accompanied by cold sweats. My hands are quaky and I feel like I just got punched in the stomach. What could this mean?

[DP-Generation (Kurakin et al., 2023)]

No filtration

Response: Hello. It sounds like you may have suffered a stroke or a mild stroke in the brain. In the past 24-48 hours I have had no speech, no facial weakness, no arm weakness, no difficulty swallowing, no speech impairment, no speech expression, no sudden severe headache with no known cause. If these symptoms happen frequently or last for a long time, it would be a good idea to talk to your doctor.

Disadvantages: Meaningless duplications and irrelevant symptoms.

[DP-Instruct (Yu et al., 2024)]

Filtration based on similarity

Response: Hi, I'm sorry to hear about your injury. I'm not a doctor, but I can give you some general advice. 1. If you have a headache, it's important to see a doctor. 2. If you feel like you just got punched in the stomach, it's important to see a doctor. 3. If you have complications such as inflammation or nerve irritation, it's important to see a doctor.

Disadvantages: Unhelpful advice.

[KnowledgeSG (Wang et al., 2024)]

Distillation from LLM to generation proxy model

Response: Given the characteristics you've described, it is crucial to have this evaluated by a healthcare professional. You should see a healthcare provider. A thorough physical examination should be conducted to assess your physiological status. Depending on the findings, the healthcare provider might recommend imaging studies to monitor your vital signs and evaluate your physiological functions.

Disadvantages: Unprofessional analyses.

[RewardDS]

Response : Hello. Welcome to Chat Doctor. The symptoms you described could be caused by several possible factors, such as stress, anxiety, infection, acid reflux, or other conditions. To determine the exact cause, I recommend scheduling an appointment with a cardiologist and undergoing diagnostic tests like blood work, an ECG, and an ultrasound. Thank you and best regards, Chat Doctor.

Advantages: Concise and helpful suggestions.

Figure 7: A representative example from the medical QA task illustrates the high quality of the response generated by *RewardDS*, compared to baseline methods. Text highlighted in red indicates meaningless or flawed parts of the answer, while text in green marks meaningful and helpful content. We also provide short analyses explaining the disadvantages or advantages of the generated responses.

G Details of Privacy Protection Evaluation

In this section, we provide further details about our attack methods, including the Data Extraction Attack (Carlini et al., 2021) and Membership Inference Attack (Yeom et al., 2018; Choquette-Choo et al., 2021). We implement both attacks for our method and the baseline methods and lower attack performance indicates stronger privacy protection capacity.

Data Extraction Attack. According to Carlini et al. (2021), the Data Extraction Attack aims to recover private fine-tuned data from the fine-tuned model. Specifically, the attackers provide the fine-tuned model with partial prefixes of private data and attempt to reconstruct the corresponding complete private data. We implement this attack on our method and the baselines to evaluate their privacy protection capabilities. The implementation details are as follows:

In our scenario, the private data consists of two components: the user query and corresponding answer, both of which may contain sensitive information. We apply the data extraction attack twice to recover the user query and the answer separately. For the user query, we provide the generation proxy model with the first 10 tokens of the private query

and prompt it to reconstruct the complete query. Then, to extract the answer, we provide the model with the previously recovered user query and the first 10 tokens of the private answer, prompting it to generate the full private response. We use greedy decoding during generation and set the maximum output length to 256.

To evaluate the attack's performance, we utilize the ROUGE-L score between the recovered data (user query and answer) and ground true private data. A higher ROUGE-L score indicates better attack performance.

Membership Inference Attack: As proposed by Yeom et al. (2018); Choquette-Choo et al. (2021), the membership inference attack aims to determine whether a specific data point was included in the private dataset used for fine-tuning. Specifically, the attackers collect numerous mixed data, which may contain some private data, and utilize the fine-tuned model to judge which one is included in the private data. We implement this attack on both our method and the baselines to assess their privacy protection capabilities. The implementation details are as follows:

To construct the mixed dataset, we apply data augmentation techniques, such as synonym replacement and content rewriting, to the private data and generate synthetic samples that are similar in con-

tent but not identical to the original private data. **For *RewardDS***, we input the mixed data into the reward proxy model and obtain the corresponding reward scores. Samples with higher reward scores are considered more likely to be part of the private training data. **For the baseline methods** (DP-Generation, DP-Instruct, and KnowledgeSG), only the generation proxy model is transferred to the server. We then use this model to compute the Perplexity (PPL) of each sample in the mixed dataset. Samples with lower PPL values are considered as the private data.

To evaluate the effectiveness of the attack, we calculate the F1 score of private data identification. A higher F1 score indicates stronger attack performance and thus weaker privacy protection.

H More Analysis of *RewardDS* Design

H.1 The impact of different privacy budget allocations.

As described in Section 6.1, we allocate an equal privacy budget to generation proxy model training and reward proxy model training. To explore the impact of different privacy budget allocations, we vary the privacy budget allocation while keeping the total privacy budget fixed at $(16, 2e^{-5})$.

As shown in Figure 8, our method performs consistently well across various allocations, except in the extreme case where no budget is allocated to reward model training (i.e., "16+0"). No budget for reward model means that we do not train the reward proxy model on the client for data filtering or refinement. This phenomenon demonstrates the critical role of the reward model. Notably, even allocating a small budget to the reward model (e.g., "15+1") leads to a significant performance boost over the "16+0" case, suggesting that **even a minimal privacy cost for reward model training yields substantial benefits**.

H.2 Generalizability across more LLM backbones.

We have evaluated our *RewardDS* on more LLM backbones, such as Llama-2-7B-chat-hf (MetaAI, 2023) and Qwen2.5-14B-Instruct (Yang et al., 2024b). Due to the computational resource constraints, we conduct the full-parameter fine-tuning for Llama-2-7B-chat-hf on the synthetic data and apply the LoRA fine-tuning (Hu et al., 2022) for Qwen2.5-14B-Instruct. We set the lora rank r as 64 and α at 16. We add the lora layer for each linear

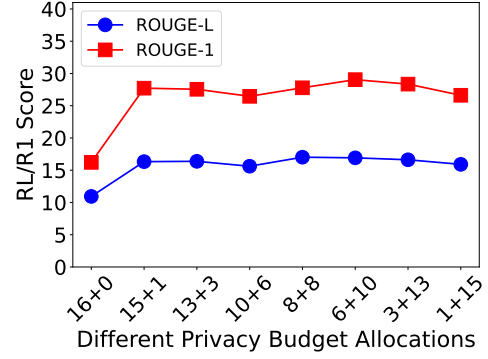


Figure 8: Performance on medical QA with different privacy budget allocations for generation proxy model and reward proxy model training. The allocation of 'x + (16-x)' means the privacy budget for training the generation proxy model is set to x, while the reward proxy model is set to (16-x);

layer in the Qwen2.5-14B-Instruct model.

As shown in Table 8, *RewardDS* outperforms other baselines regardless of whether Llama-2-7B-chat-hf or Qwen2.5-14B-Instruct is used as the LLM backbone. This strongly demonstrates that our method is consistently effective, regardless of the LLM backbone. It is worth noting that although Qwen2.5-14B-Instruct has a larger number of parameters compared to Llama-2-7B-chat-hf, our method performs better on the Llama-2-7B-chat-hf model. This is likely due to the use of LoRA fine-tuning on Qwen2.5-14B-Instruct, rather than full-parameter fine-tuning. We believe that applying full-parameter fine-tuning to the Qwen2.5-14B-Instruct model would lead to better performance. Overall, **our method consistently achieves superior performance across various LLM backbones, which strongly demonstrates its generalizability**.

I More Hyperparameter Analysis

In this section, we analyze the other hyperparameters of our method, including the number of folds k and the number of candidate responses N , for the medical QA, financial QA and code generation tasks.

As described in Alg. 1, **the number of folds k** controls how much of the synthetic data is considered clean. As shown in Figure 9, $k = 6$ yields the best performance on the medical QA task. For the financial QA and code generation tasks, the optimal values are $k = 5$ (Figure 10) and $k = 8$ (Figure 11), respectively. Larger k values lead to stricter filtering, excluding more data, which may cause overfitting on smaller subsets and degrade